Chapter 9: Python GUI

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Time required: 120 minutes

DRY

Don't **R**epeat **Y**ourself

Tutorials

- https://realpython.com/python-gui-tkinter/

Python GUI - Tkinter

Up until now, the only way our programs have been able to interact with the user is through keyboard input via the input statement. These types of programs determine the order in which things happen.

GUI programs are event driven; the user is in control. The user can click or enter data in any order. Mouse clicks or keyboard events are registered: the program responds to that event with the appropriate action.

Graphical programs use windows, buttons, scrollbars, and various other things. These widgets are part of what is called a Graphical User Interface or GUI.

This chapter is about GUI programming in Python with **Tkinter**. GUI programming is not built into Python. Python comes with a built-in module called **tkinter** that you can import.

There are a couple approaches to using **tkinter**. This chapter will take the object-oriented approach, as is more flexible and reusable. Although tktinter code can be written using only functions, it's much better to use a class to keep track of all individual widgets which may need to reference each other. Without doing this, you need to rely on global or nonlocal variables, which gets ugly as your app grows and is not a secure method of programming. It allows for

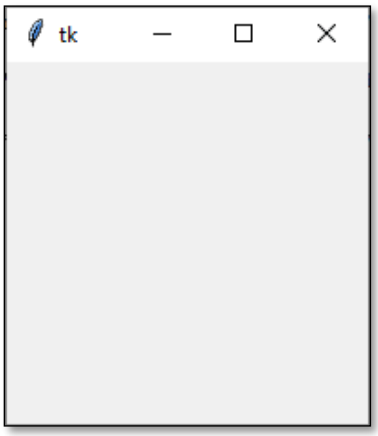
much finer controls once your app gets more complex, allowing you to override default behaviors of tinder's own objects.

A GUI program starts with a window with various widgets for the user to interact with. The widgets we will be looking at have far more options than we could possibly cover.

Tutorial 9.1: Hello Tkinter World!

Let's start out with the minimum code needed to create an empty window.

```
1   """
2       Name: window_empty_simple.py
3       Author:
4       Created:
5       Purpose: Display an empty window in 3 lines of code
6   """
7
8   # Import the tkinter module
9   from tkinter import *
10
11  # Create the main window
12  root = Tk()
13
14  # Loop the program to keep displaying the window
15  root.mainloop()
```



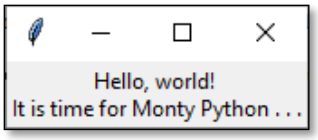
Not very impressive, is it. The code behind it isn't very complicated either. Let's modify this program into the traditional **Hello World** program. We create a root window, put a title in the title bar, and display a single label.

We will program in Tkinter using OOP. This is a common practice in Python. It allows more flexibility as we learn more about Python.

Create a python program named **hello_world.py**

```python
"""
    Name: hello_world.py
    Author:
    Created:
    Purpose: Traditional Hello World program
"""
# Import the tkinter module
from tkinter import *


class HelloWorld:
    # Define the initialize method
    def __init__(self):
        # Create the root window
        self.root = Tk()

        # Title bar to the root window
        self.root.title("Hi")

        # Create a label
        self.lbl_display = Label(
            self.root,
            text="Hello, World! \nIt is time for Monty Python . . ."
        )

        # Pack the label, pack is a window/layout manager
        # Pack makes the widget visible in the window
        self.lbl_display.pack()

        # Call the mainloop method which is used
        # when the application is ready to run
        # It tells the application to keep displaying the GUI
        mainloop()


# Create program object
hello_world = HelloWorld()
```

Example run:



How It Works

```
# Import the tkinter module
from tkinter import *
```

This line imports the **tkinter** module. It contains all classes, functions and other items needed to work with the **Tk** GUI toolkit.

```
# Create program object
hello_world = HelloWorld()
```

The program starts by creating an object from the program class.

```
class HelloWorld:
    # Define the initialize method
    def __init__(self):
        # Create the root window
        self.root = Tk()
```

After the object is created, we go to the dunder **__init__** method. To initialize **tkinter**, we create a root window. This is an ordinary window, with a title bar and other decorations provided by your window manager. Create only one root window for each program. It must be created before any other widgets.

```
# Create a label
self.lbl_display = Label(self.root,
    text = "Hello, world! \nIt is time for Monty Python . . .")
```

We create a Label widget named **lbl_display** as a child to the root window:

A **Label** widget can display either text, an icon or other image. In this case, we use the text option to specify which text to display.

```
# Pack the label
# Size it to fit the text
# Make it visible
self.lbl_display.pack()
```

We call the **pack** method on this widget. This tells it to size itself to fit the given text and make itself visible. The window and other GUI elements won't appear until we've entered the **tkinter mainloop()** event loop:

```
# Call the mainloop method which is used
# when the application is ready to run
# It tells the application to keep displaying the GUI
mainloop()
```

The program will stay in the event loop until we close the window. The event loop handles events from the user (such as mouse clicks and key presses) and the windowing system (such as redraw events and window configuration messages). It also handles operations queued by **tkinter** itself. Among these operations are geometry management (queued by the pack method) and display updates. The application window will not appear before you enter the main loop.

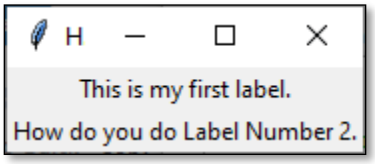
Tutorial 9.2: Labels

This program is very similar to the last program: we add a second label. The **pack** method must be run on each label and ensures that each label is the correct size and is visible.

NOTE: Some program examples will use window, some will use root. Either one is fine. window and root are conventions.

```python
"""
    Name: window_2_labels.py
    Author:
    Created:
    Purpose: Display 2 labels with OOP
"""
# Import the tkinter module
from tkinter import *

# Define the class
class Window2Labels:
    # Define the initialize method
    def __init__(self):
        # Create the root window
        self.root = Tk()

        # Title bar on the root window
        self.root.title("Hi")

        # Create 2 label widgets
        self.lbl_display = Label(
            self.root,
            text="This is my first label.",
        )
        self.lbl_display2 = Label(
            self.root,
            text="How do you do Label Number 2."
        )

        # Pack both labels to the window
        self.lbl_display.pack()
        self.lbl_display2.pack()

        # Start the application
        mainloop()


# Create an instance/object from the program class
window_2_labels = Window2Labels()
```

Example run:



We can change the alignment of our widgets by passing arguments to the **pack** method.

```
# Pack both labels aligned to their left side
# Size them to fit the text
# Make themselves visible
self.lbl_display.pack(side='left')
self.lbl_display2.pack(side='left')
```

Example run:

The other arguments for **side= top**, **bottom**, **left**, and **right**.

Labels

Label Options: There are several options you can change including font size and color. Here are some examples:

```
hello_label = Label(text='hello', font=('Verdana', 24, 'bold'),
                    bg='blue', fg='white')
```

Note the use of keyword arguments. Here are a few common options:

- **font**: The basic structure is **font=(font name, font size, style)**. You can leave out the font size or the style. The choices for style are 'bold', 'italic', 'underline', 'overstrike', 'roman', and 'normal' (which is the default). You can combine multiple styles like this: 'bold italic'.
- **foreground and background:** These stand for foreground and background colors. Many common color names can be used, like 'blue', 'green', etc.
- **width**: This is how many characters long the label should be. If you leave this out, Tkinter will base the width off the text you put in the label. This can make for

unpredictable results. It is good to decide ahead of time how long you want your label to be and set the width accordingly.

- **Height**: This is how many rows high the label should be. You can use this for multiline labels. Use newline characters in the text to get it to span multiple lines.

For example:

text='hi\nthere'

Example run:

Hi
there

Borders and Padding

You can also display a border around a label and use internal and external padding. Each method can have more than one argument separated by commas.

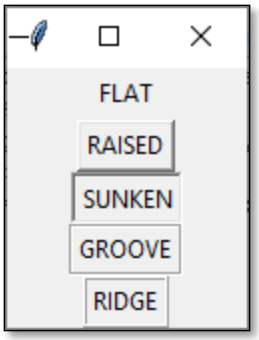
Two border arguments are

borderwidth = 1, relief = 'solid')

borderwidth can have an integer value which specifies the width in pixels.

relief can have **FLAT, RAISED, SUNKEN, RIDGE, SOLID, and GROOVE**.

```
from tkinter import *
root = Tk()

button_1 = Button(root, text="FLAT", relief=FLAT)
button_2 = Button(root, text="RAISED", relief=RAISED)
button_3 = Button(root, text="SUNKEN", relief=SUNKEN)
button_4 = Button(root, text="GROOVE", relief=GROOVE)
button_5 = Button(root, text="RIDGE", relief=RIDGE)

button_1.pack()
button_2.pack()
button_3.pack()
button_4.pack()
button_5.pack()
root.mainloop()
```



Label Options

anchor	It specifies the exact position of the text within the size provided to the widget. The default value is CENTER, which is used to center the text within the specified space.
bg	The background color displayed behind the widget.
bitmap	It is used to set the bitmap to the graphical object specified so that, the label can represent the graphics instead of text.
bd	It represents the width of the border. The default is 2 pixels.
cursor	The mouse pointer will be changed to the type of the cursor specified, i.e., arrow, dot, etc.
font	The font type of the text written inside the widget.
fg	The foreground color of the text written inside the widget.
height	The height of the widget.
image	The image that is to be shown as the label.
justify	It is used to represent the orientation of the text if the text contains multiple lines. It can be set to LEFT for left justification, RIGHT for right justification, and CENTER for center justification.
padx	The horizontal padding of the text. The default value is 1.
pady	The vertical padding of the text. The default value is 1.

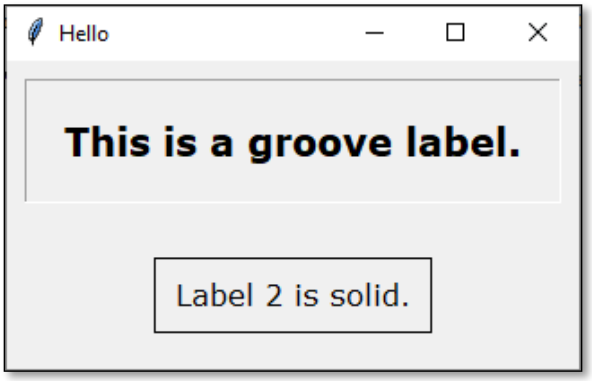
relief	The type of the border. The default value is FLAT. Options are: GROOVE, SOLID, SUNKEN, RIDGE
text	This is set to the string variable which may contain one or more line of text.
textvariable	The text written inside the widget is set to the control variable StringVar so that it can be accessed and changed accordingly.
underline	We can display a line under the specified letter of the text. Set this option to the number of the letter under which the line will be displayed.
width	The width of the widget. It is specified as the number of characters.
wraplength	Instead of having only one line as the label text, we can break it to the number of lines where each line has the number of characters specified to this option.

Tutorial 9.3: Label Options

This tutorial combines some of the methods and properties we have been looking at.

```python
"""
    Name: window_padding.py
    Author:
    Created:
    Display 2 labels with borders, padding and other options
"""

# Import the tkinter module
from tkinter import *


class WindowBordersPadding:
    # Define the initialize method
    def __init__(self):
        # Create the root window
        self.root = Tk()

        # Title bar to the root window
        self.root.title("Hello")

        # Create a label
        self.lbl_display = Label(
            self.root,
            text="This is a groove label.",
            borderwidth=1,
            relief=GROOVE,
            font=('Verdana', 16, 'bold')
        )
        self.lbl_display2 = Label(
            self.root,
            text="Label 2 is solid.",
            borderwidth=1,
            relief=SOLID,
            font=('Verdana', 12)
        )

        # pack layout manager displays widgets in the window
        self.lbl_display.pack(ipadx=20, ipady=20, padx=10, pady=10)
        self.lbl_display2.pack(ipadx=10, ipady=10, padx=20, pady=20)

        # Call the mainloop method runs the program
        mainloop()


# Create an instance/object from the program class
window_borders_padding = WindowBordersPadding()
```

Example run:



Changing Label Properties

After you've created a label or any other widget, you may want to change something about it. To do that, use the **configure** method. Here are two examples that change the properties of a **label** called **label**:

```
label.configure(text='Bye')
label.configure(bg='white', fg='black')
```

Setting text to something using the **configure** method is kind of like the GUI equivalent of a **print** statement. In calls to **configure** we cannot use commas to separate multiple things to print. We instead need to use **string** formatting. Here is a **print** statement and its equivalent using the **configure** method.

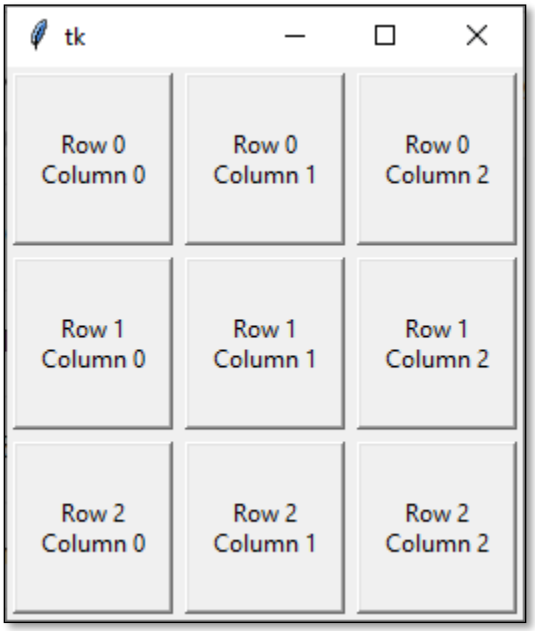
```
print('a =', a, 'and b =', b)
label.configure(text='a = {}, and b = {}'.format(a,b))
```

Grid Layout Manager

The **grid** method is used to place things on the screen. It lays out the screen as a rectangular grid of rows and columns. The first few rows and columns are shown below.

(row=0, column=0)	(row=0, column=1)	(row=0, column=2)
(row=1, column=0)	(row=1, column=1)	(row=1, column=2)
(row=2, column=0)	(row=2, column=1)	(row=2, column=2)

```
from tkinter import *
window = Tk()

for i in range(3):
    window.columnconfigure(i, weight=1)
    window.rowconfigure(i, weight=1)

    for j in range(3):
        frame = Frame(
            master=window,
            borderwidth=1
        )
        frame.grid(row=i, column=j, padx=1, pady=1)
        button = Button(
            master=frame,
            text=f"Row {i}\nColumn {j}",
            height=5,
            width=10
        )
        button.pack(padx=1, pady=1)
window.mainloop()
```



With multiple rows or columns there are optional arguments, **rowspan** and **columnspan**, that allow a widget to take up more than one row or column. Here is an example of several grid statements followed by what the layout will look like:

```
label1.grid(row=0, column=0)
label2.grid(row=0, column=1)
label3.grid(row=1, column=0, columnspan=2)
label4.grid(row=1, column=2)
label5.grid(row=2, column=2)
```

label1	label2	
label 3		label4
		label5

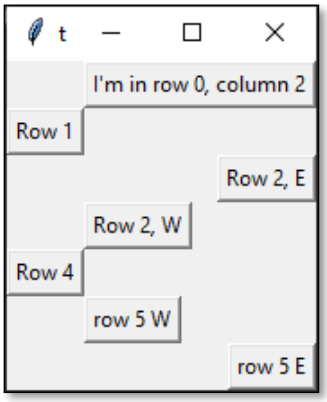
Spacing: To add extra space between widgets, there are optional arguments as follows.

- **ipadx:** How many pixels to pad widget horizontally inside the widget's borders.
- **ipady:** How many pixels to pad widget vertically inside the widget's borders.
- **padx:** How many pixels to pad widget horizontally outside the widget's borders.
- **pady:** How many pixels to pad widget vertically outside the widget's borders.

Sticky: When the widget is smaller than the cell, sticky is used to indicate which sides and corners of the cell the widget sticks to. The direction is defined by compass directions: N, E, S, W, NE, NW, SE, and SW and zero. These could be a string concatenation, for example, NESW make the widget take up the full area of the cell.

```
from tkinter import *

def main():

    root = Tk()
    btn_column = Button(root, text="I'm in row 0, column 2")
    btn_column.grid(row=0, column=1)

    btn_columnspan = Button(root, text="Row 1")
    btn_columnspan.grid(row=1, column=0)

    btn_pady = Button(root, text="Row 2, E")
    btn_pady.grid(row=2, column=1, sticky=E)

    btn_row = Button(root, text="Row 2, W")
    btn_row.grid(row=3, column=1, sticky=W)

    btn_rowspan = Button(root, text="Row 4")
    btn_rowspan.grid(row=4, column=0)

    btn_sticky = Button(root, text="row 5 W")
    btn_sticky.grid(row=5, column=1, sticky=W)

    btn_sticky1 = Button(root, text="row 5 E")
    btn_sticky1.grid(row=6, column=1, sticky=E)

    root.mainloop()
main()
```

Example run:



Important Note: Any time you create a widget, to place it on the screen you need to use **grid** (or one of its cousins, like **pack**). Otherwise it will not be visible.

Entry Boxes

Entry boxes are a way for your GUI to get text input from the user. The following example creates a simple entry box and places it on the screen.

```
entry = Entry()
entry.grid(row=0, column=0)
```

ost of the same options that work with labels work with entry boxes (and most of the other widgets we will talk about). The width option is particularly helpful because the entry box will often be wider than you need.

Syntax

```
w = Entry(master, option=value, ... )
```

Getting Text

To get the text from an entry box, use the **get** method. This will return a string. If you need numerical data, use **int** or **float** to cast the string. Here is a simple example that gets text from an entry box named **entry**.

```
string_value = entry.get()
num_value = int(entry.get())
```

Deleting text: To clear an entry box, use the following:

```
entry.delete(0, END)
```

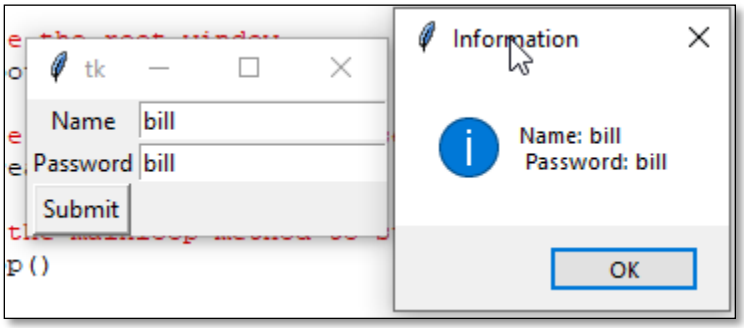
Inserting text: To insert text into an entry box, use the following:

```
entry.insert(0, 'hello')
```

Example entry program.

```
from tkinter import *
from tkinter import messagebox

class EntryDemo:
    def __init__(self):
        self.root = Tk()

        lbl_name = Label(self.root, text="Name").grid(row=0, column=0)
        self.entry_name = Entry(self.root)
        self.entry_name.grid(row=0, column=1)
        lbl_password = Label(self.root, text="Password").grid(row=1,
column=0)
        self.entry_password = Entry(self.root)
        self.entry_password.grid(row=1, column=1)
        submit = Button(
            self.root,
            text="Submit",
            command=self.get_entry,
        ).grid(row=4, column=0)
        mainloop()

    def get_entry(self):
        print(self.entry_name.get())
        name = self.entry_name.get()
        password = self.entry_password.get()
        messagebox.showinfo(
            "Information",
            f"Name: {name}\n Password: {password}"
        )

entry_demo = EntryDemo()
```

Example run:



Entry Box Options

bg	The normal background color displayed behind the label and indicator.
bd	The size of the border around the indicator. Default is 2 pixels.
cursor	If you set this option to a cursor name (*arrow, dot etc.*), the mouse cursor will change to that pattern when it is over the widget.
exportselection	By default, if you select text within an Entry widget, it is automatically exported to the clipboard. To avoid this exportation, use exportselection=0.
fg	The color used to render the text.
font	The font used for the text.
highlightcolor	The color of the focus highlight when the widget has the focus.
justify	If the text contains multiple lines, this option controls how the text is justified: CENTER, LEFT, or RIGHT.
relief	With the default value, relief=FLAT, the checkbutton does not stand out from its background. You may set this option to any of the other styles like GROOVE, RAISED, RIGID.
selectbackground	The background color to use displaying selected text.
Selectborderwidth	The width of the border to use around selected text. The default is one pixel.
selectforeground	The foreground (text) color of selected text.
show	Normally, the characters that the user types appear in the entry. To make a password entry that echoes each character as an asterisk, set show="*".
state	The default is state=NORMAL, but you can use state=DISABLED to gray out the control and make it unresponsive.

textvariable	In order to be able to retrieve the current text from your entry widget, you must set this option to an instance of the StringVar class.
width	The default width of a checkbutton is determined by the size of the displayed image or text. You can set this option to a number of characters and the checkbutton will always have room for that many characters.
xscrollcommand	If you expect that users will often enter more text than the onscreen size of the widget, you can link your entry widget to a scrollbar.

Entry Box Methods

delete (first, last=None)	Deletes characters from the widget, starting with the one at index first, up to but not including the character at position last. If the second argument is omitted, only the single character at position first is deleted.
get()	Returns the entry's current text as a string.
icursor(index)	Set the insertion cursor just before the character at the given index.
index(index)	Shift the contents of the entry so that the character at the given index is the leftmost visible character. Has no effect if the text fits entirely within the entry.
insert(index, s)	Inserts string s before the character at the given index.
select_adjust (index)	This method is used to make sure that the selection includes the character at the specified index.
select_clear()	Clears the selection. If there isn't currently a selection, has no effect.
select_from (index)	Sets the ANCHOR index position to the character selected by index, and selects that character.
select_present()	If there is a selection, returns true, else returns false.

select_range (start, end)	Sets the selection under program control. Selects the text starting at the start index, up to but not including the character at the end index. The start position must be before the end position.
select_to(index)	Selects all the text from the ANCHOR position up to but not including the character at the given index.
xview (index)	This method is useful in linking the Entry widget to a horizontal scrollbar.
xview_scroll (number, what)	Used to scroll the entry horizontally. The what argument must be either UNITS, to scroll by character widths, or PAGES, to scroll by chunks the size of the entry widget. The number is positive to scroll left to right, negative to scroll right to left.

Buttons

The Button widget is used to add buttons in a Python application. These buttons can display text or images that convey the purpose of the buttons. You can attach a function or a method to a button which is called automatically when you click the button.

Syntax to create this widget.

```
w = Button(master, option=value, ... )
```

The following example creates a simple button:

```
btn_ok = Button(text='Ok')
```

The follow options can be used with the Python Button object.

activebackground	Background color when the button is under the cursor.
activeforeground	Foreground color when the button is under the cursor.
bd	Border width in pixels. Default is 2.
bg	Normal background color.
command	Function or method to be called when the button is clicked.

fg	Normal foreground (text) color.
font	Text font to be used for the button's label.
height	Height of the button in text lines (for textual buttons) or pixels (for images).
highlightcolor	The color of the focus highlight when the widget has focus.
image	Image to be displayed on the button (instead of text).
justify	How to show multiple text lines: LEFT to left-justify each line; CENTER to center them; or RIGHT to right-justify.
padx	Additional padding left and right of the text.
pady	Additional padding above and below the text.
relief	Relief specifies the type of the border. Some of the values are SUNKEN, RAISED, GROOVE, and RIDGE.
state	Set this option to DISABLED to gray out the button and make it unresponsive. Has the value ACTIVE when the mouse is over it. Default is NORMAL.
underline	Default is -1, meaning that no character of the text on the button will be underlined. If nonnegative, the corresponding text character will be underlined.
width	Width of the button in letters (if displaying text) or pixels (if displaying an image).
wraplength	If this value is set to a positive number, the text lines will be wrapped to fit within this length.

To get the button to do something when clicked, use the **command** argument. It is set to the name of a method, called a **callback** method. When the button is clicked, the **callback** method is called.

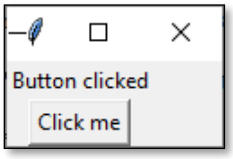
Tutorial 9.4: Buttons

Create and save the following program as **callback_demo.py**

When the program starts, the label says **Click me**. When the button is clicked, the **callback** method is called, which changes the label to say Button clicked. A **callback** method can be named anything, we just choose callback for this example. If you have other buttons, you can create other methods for them.

```python
"""
    Name: callback_demo.py
    Author:
    Created:
    Purpose: Demonstrate handling a button click event
"""

# Import the tkinter module
from tkinter import *


class CallBackDemo:
    # Define the initialize method
    def __init__(self):
        # Create the root window
        self.root = Tk()

        # Create the GUI widgets in a separate method
        self.create_widgets()

        # Keep displaying our GUI
        mainloop()

    def display_text(self):
        """Callback method to handle the button click event"""
        self.label.configure(text="Button Clicked")

    def create_widgets(self):
        """Define the create widgets method"""
        self.label = Label(self.root, text="Not Clicked")

        # Define the button and button click event command action
        self.button = Button(
            self.root,
            text="Click Me",
            command=self.display_text
        )

        # Align the widgets to display on the window
        self.label.grid(row=0, column=0)
        self.button.grid(row=1, column=0)


# Create an instance/object from the program class
call_back_demo = CallBackDemo()
```

Example run:



Colors

Tkinter defines many common color names, like 'yellow' and 'red'.

```
label = Label(text = 'Hi', background = 'blue')
```

There are color names built into **tkinter**. The **display_colors.py** attached to this assignment will display all the built-in color names. The following is an example.

```
label = Label(text = 'Hi', background = 'cyan')
```

Ttk Themed Widgets

The tkinter.ttk module provides access to the Tk themed widget set, added in 2007 with Tk 8.5. Ttk stands for Tk themed.

To override the basic Tk widgets, the Ttk import should follow the Tk import:

```
from tkinter import *
from tkinter.ttk import *
```

This code causes several tkinter.ttk widgets (Button, Checkbutton, Entry, Frame, Label, LabelFrame, Menubutton, PanedWindow, Radiobutton, Scale and Scrollbar) to automatically replace the Tk widgets.

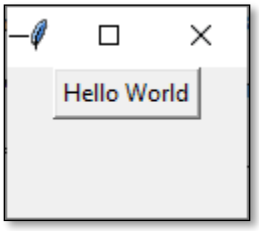
The new widgets give a better look and feel across platforms. Tthe replacement widget optioins are not completely compatible. The main difference is that widget options such as “fg”, “bg” and others related to widget styling are no longer present in Ttk widgets. Instead, you would use the ttk.Style class for improved styling effects.

Ttk Widgets

Ttk comes with 18 widgets, twelve of which already existed in tkinter: Button, Checkbutton, Entry, Frame, Label, LabelFrame, Menubutton, PanedWindow, Radiobutton, Scale, Scrollbar, and Spinbox. The other six are new: Combobox, Notebook, Progressbar, Separator, Sizegrip and Treeview.

Using the Ttk widgets give the application an improved look and feel. There are differences in how the styling is coded.

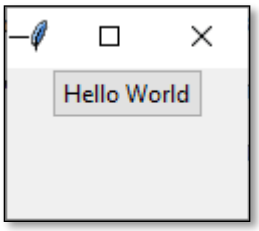
Tk code for Hello World:




```
from tkinter import *
root = Tk()
Button(root, text="Hello World").grid()
mainloop()
```

Ttk code for Hello World:

E:_wncc\Python\Assignments\13 Math\Circes Circle Calculator\circle_calculator_functions



_oop.py

```
from tkinter import *
from tkinter.ttk import *
root = Tk()
Button(root, text="Hello World").grid()
mainloop()
```

Tutorial 9.5: Temperature Converter

Let's create a tkinter ttk version of our old friend, the temperature converter.

Find a free thermometer icon. The extension for Windows icon files is ico.
IconArchive (Free ico files)

If you wish to put in a degree symbol,:

1. https://www.degreesymbol.net/
2. Click the Copy button.

3. Paste in your code. °

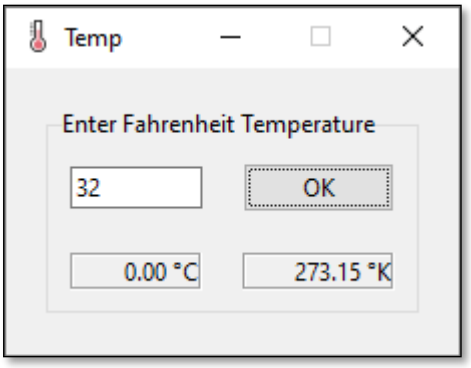
Create a Python program named: **temperature_converter_gui.py**

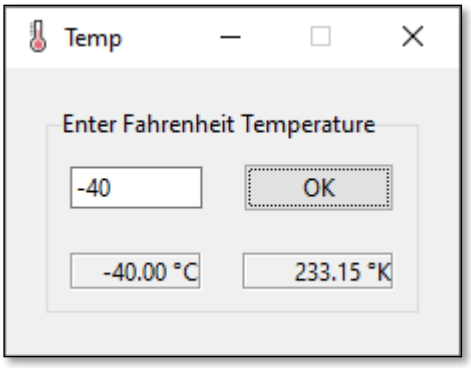
```python
"""
    Name: temperature_converter_gui.py
    Author:
    Created:
    Purpose: Convert Fahrenheit to Celsius and Kelvin
"""
# Import the tkinter module with tk standard widgets
from tkinter import *
# Override tk widgets with themed ttk widgets if available
from tkinter.ttk import *


class TemperatureConverter:
    def __init__(self):
        # Create the root window
        self.root = Tk()
        self.root.title("Temp")
        # Add icon to window corner
        # Search the web for free thermometer.ico files
        self.root.iconbitmap("thermometer.ico")
        # Prevent window from resizing
        self.root.resizable(False, False)
        # Create the GUI widgets in a separate method
        self.create_widgets()
        # Call the mainloop method to start program
        mainloop()

# ---------------------- CONVERT TEMPERATURE ----------------------------#
    def convert_temperature(self, *args):
        """Method called by the button click event
           Convert fahrenheit to celsius and kelvin.
        """
        # Get input from user as a float
        self.fahrenheit = float(self.entry.get())

        # Convert Fahrenheit to Celsius and Kelvin
        self.celsius = ((self.fahrenheit - 32) * 5.0) / 9.0
        self.kelvin = (((self.fahrenheit - 32) * 5.0) / 9.0) + 273.15

        # Display output in labels using the configure method
        self.lbl_celsius.configure(text=f"{self.celsius:,.2f} °C")
        self.lbl_kelvin.configure(text=f"{self.kelvin:,.2f} °K")

        # 0 starts the selection at the beginning of the entry widget text
        # END finishes the selection at the end of the entry widget text
        self.entry.selection_range(0, END)
```

```python
# ---------------------- CREATE WIDGETS ---------------------------------#
    def create_widgets(self):
        """Create and grid widgets."""
        # Create main label frame to hold widgets
        self.main_frame = LabelFrame(
            self.root,                          # Assign to parent window
            text="Enter Fahrenheit Temperature",  # Text for the frame
            relief=GROOVE                       # Decorative border
        )

        # Create entry widget in the frame to get input from user
        self.entry = Entry(
            self.main_frame,  # Assign to parent frame
            width=10          # Width in characters
        )

        # Create button in the frame to call calulate method
        self.btn_calculate = Button(
            self.main_frame,     # Assign to parent frame
            text="OK",           # Text shown on button
            # Connect convert method to button click
            command=self.convert_temperature
        )

        # Create label in the frame to show celsius
        self.lbl_celsius = Label(
            self.main_frame,     # Assign to parent frame
            width=10,            # Width in characters
            relief=GROOVE,       # Decorative border
            anchor=E             # Anchor text to the East
        )

        # Create label in the frame to show output
        self.lbl_kelvin = Label(
            self.main_frame,
            width=10,
            relief=GROOVE,
            anchor=E
        )
```

```python
        # Use Grid layout manager to place widgets in the frame
        self.entry.grid(row=0, column=0)
        self.btn_calculate.grid(row=0, column=1)

        # sticky uses cardinal directions to stick labels
        # to sides of the column
        # sticky=EW expands the label to fit the column
        # based on the largest widget
        self.lbl_celsius.grid(
            row=1,
            column=0,
            sticky=EW
        )
        self.lbl_kelvin.grid(
            row=1,
            column=1,
            sticky=EW
        )

        # Set padding between frame and window
        self.main_frame.grid_configure(padx=20, pady=20)

        # Set padding for all widgets inside the frame
        for widget in self.main_frame.winfo_children():
            widget.grid_configure(padx=10, pady=10)

        # Start the program with focus on the entry widget
        self.entry.focus_set()

        # Bind both enter key to the convert method
        # When either Enter key is pressed,
        # the convert method will be fired
        # <Return> - Enter key on main keyboard
        # <KP_Enter> - Enter key on number pad/key pad
        self.root.bind("<Return>", self.convert_temperature)
        self.root.bind("<KP_Enter>", self.convert_temperature)


# -------------------- MAIN PROGRAM -------------------------------------#
"""Create program object from the program class to run the program."""
temperature_converter = TemperatureConverter()
```

Example run:





Using tkinter, ttk, and Frames in Temperature Converter

In temperature_converter.py, we added ttk to tkinter and frames to control the layout.

Our program starts by incorporating Tk as we have done in the past.

```
from tkinter import *
from tkinter.ttk import *
```

These import two built-in Python modules. **tkinter** is the standard Python binding to the Tk GUI library. When imported, it loads the Tk library on your system. **ttk** is a submodule of **tkinter**. It implements Python's binding to the newer "themed widgets" that were added to Tk in 8.5.

NOTE: One gotcha to watch out for, the newer ttk widgets do not always have the same options as the tk widgets.

Several widgets are defined in both modules. By importing everything from ttk, we don’t have to know which is which. If there is not a newer ttk widget, the standard tkinter widget will be used.

Instead of placing the widgets inside the window, we placed them inside a nice label frame with a border. You can add more than one frame, allowing you to divide up the purpose of various parts of your program.

The other reason is that the main window isn't itself part of the newer "themed" widgets. Its background color doesn't match the themed widgets we will put inside it. Using a "themed" frame widget to hold the content ensures that the background color of the widget is correct.

Using Frames and Widgets with OOP

When using a frame to place our widgets, we need to do two things: create the widget itself and then place it onscreen.

When we create a widget, we need to specify its *parent*. That is the widget that the new widget will be placed inside. In this case, we want our entry placed inside the content frame. Our entry, and other widgets we'll create shortly, are said to be *children* of the content frame. In Python, the `parent` is passed as the first parameter when instantiating a widget object as shown below.

```
        # Create entry widget in the frame to get input from user
        self.entry = Entry(
            self.main_frame,
            text='',
            width=10)
```

When we create a widget, we can provide it with certain *configuration options*. In the example above, we specify how wide we want the entry to appear, i.e., 10 characters.

When widgets are created, they don't automatically appear on the screen; Tk doesn't know where you want them placed relative to other widgets. That's what the `grid` part does. Widgets are placed in the appropriate column (0 or 1) and row (also 0 or 1).

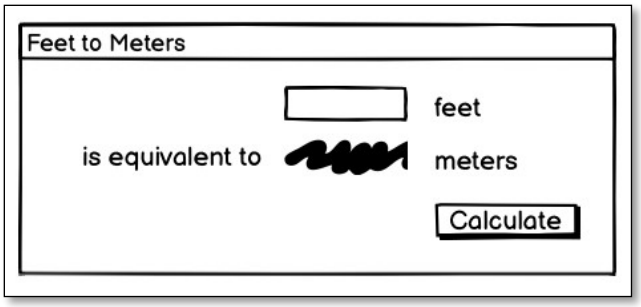
```
        # Use Grid layout manager to place widgets in the frame
        self.entry.grid(row=0, column=0)
        self.btn_calculate.grid(row=0, column=1)
        self.lbl_celsius.grid(row=1, column=0)
        self.lbl_kelvin.grid(row=1, column=1, sticky=E)
```

The `sticky` option to grid describes how the widget should line up within the grid cell, using compass directions. E (east) means to anchor the widget to the right side of the cell, WE (west-east) anchors the widget to both the left and right sides, and so on. Python defines constants for these directional strings, which you can provide as a list, e.g., `W` or `(W, E)`.

GUI Design with Pencil

Say we want to create a GUI for a meters to feet calculator.

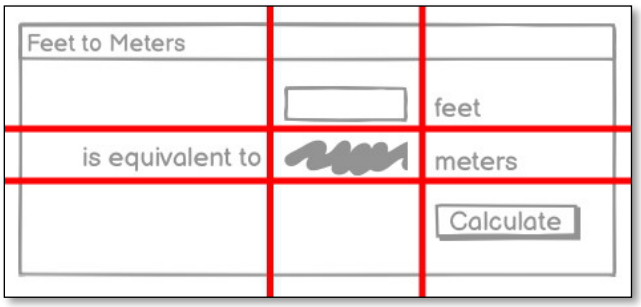
Sketch your interface on paper. It might look something like this:



Here are the widgets we will need.

1. Text entry to type in the number of feet.
2. Calculate button will get the value out of that entry, perform the calculation, and put the result in a label below the entry.
3. Three static labels ("feet," "is equivalent to," and "meters"), which help our user figure out how to run the application.

Let’s look at the layout. The grid layout is very flexible for this type of GUI design. The widgets we've included seem to be naturally divided into a grid with three columns and three rows. In terms of layout, things seem to naturally divide into three columns and three rows, as illustrated below:



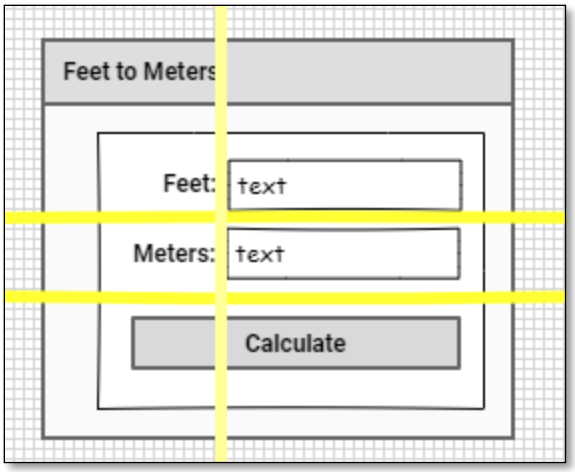
The layout of our user interface with a 3 x 3 grid.

Pencil

Pencil is a free and open-source GUI prototyping tool that people can easily install and use to create mockups in popular desktop platforms. This type of design is called a wireframe.

1. Go to https://pencil.evolus.vn/
2. Download and install Pencil for your OS.

Example wireframe of our evolving GUI:



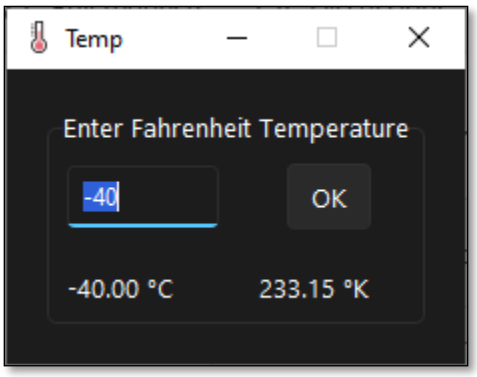
This example uses a grid layout. 2 columns, 3 rows inside a frame.

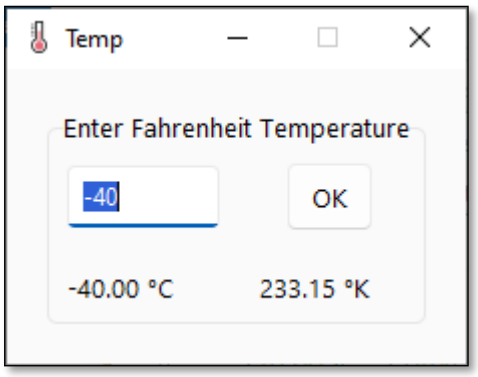
Tutorial 9.6: Temperature with Sun Valley Theme

Tkinter isn't a very modern looking interface. One of the most interesting things about Python is how many open-source libraries can be added to Python to solve almost any problem.

This Tkinter themed example is created from rdbende's Sun Valley ttk theme library.

https://github.com/rdbende/Sun-Valley-ttk-theme





Temperature Converter Theme Code Changes

Save **temperature_converter.py** as **temperature_converter_theme.py**

```python
"""
    Name: temperature_converter_theme.py
    Author:
    Created:
    Purpose: Convert Fahrenheit to Celsius and Kelvin
    https://github.com/rdbende/Sun-Valley-ttk-theme
"""
# Import the tkinter module with tk standard widgets
from tkinter import *
# Override tk widgets with themed ttk widgets if available
from tkinter.ttk import *
# pip install sv-ttk
# Import Sun Valley Theme
import sv_ttk
```

1. Install the sv-ttk library: **pip install sv-ttk**
2. Import the new library: **import sv_ttk**

```python
        # Create the GUI widgets in a separate method
        self.create_widgets()

        # Set the theme to light or dark
        sv_ttk.set_theme("light")

        # Call the mainloop method to start program
        mainloop()
```

3. Set the theme to light or dark as shown. The above example is light.

Tutorial 9.7: ttkbootstrap

ttkbootstrap is a supercharged theme extension for tkinter that enables on-demand modern flat style themes. https://ttkbootstrap.readthedocs.io/en/latest/

ttkbootstrap is used almost exactly like Tkinter, the only difference is a slight change in the __init__ method. It has pre-defined themes and styles, new widgets, and a theme creator.

Try some of the other themes. This example uses darkly.

https://ttkbootstrap.readthedocs.io/en/latest/themes/

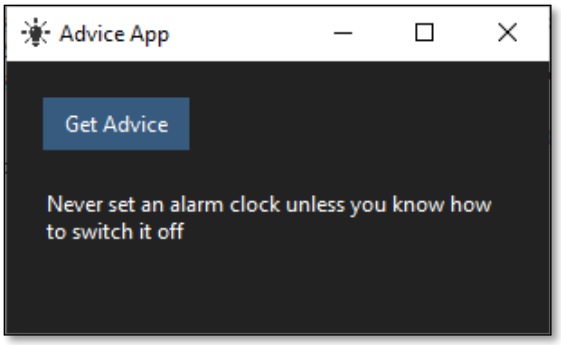
Install ttkbootstrap.

```
pip install ttkbootstrap
# Demo styles program: python -m ttkbootstrap
```

```python
"""
    Name: advice_gui.py
    Author: William Loring
    Created: 01/07/2023
    Purpose: Get random advice from API using ttkbootstrap
"""
# pip install ttkbootstrap
# Override tk widgets with ttkbootstrap
import ttkbootstrap as ttk
from ttkbootstrap.constants import *
import requests

# Set this to False to only display the result
IS_DEBUGGING = False

# URL for single random advice
URL = "https://api.adviceslip.com/advice"


class AdviceGUI:
    def __init__(self):
        self.root = ttk.Window(themename="darkly")
        self.root.title("Advice App")
        self.root.minsize(width=300, height=150)
        # Padding to edge of window
        self.root.config(padx=10, pady=10)
        self.root.iconbitmap("idea.ico")
        self.create_widgets()
        self.root.mainloop()

# ------------------------------ CREATE WIDGETS -------------------------#
    def create_widgets(self):
        self.btn_get_advice = ttk.Button(
            text="Get Advice",
            command=self.get_data
        )
        self.lbl_advice = ttk.Label(
            wraplength=250,
            justify="left",
        )
        PAD = 10
        self.btn_get_advice.grid(row=0, column=0, padx=PAD, pady=PAD, sticky=W)
        self.lbl_advice.grid(row=1, column=0, padx=PAD, pady=PAD, sticky=W)

        # The enter key will activate the calculate method
        self.root.bind("<Return>", self.get_data)
        self.root.bind("<KP_Enter>", self.get_data)
        self.root.bind("<Escape>", self.close_window)
```

```python
# ------------------------------- GET DATA -------------------------------#
    def get_data(self, *args):
        # Use the requests.get() function to get data from API
        response = requests.get(URL)
        # Watch for exception
        response.raise_for_status()

        # If the status_code is 200, successful connection and data
        if (response.status_code == 200):

            # Convert JSON data into Python dictionary with key value pairs
            self._advice_data = response.json()

            # Used to debug process
            if (IS_DEBUGGING == True):

                # Display the status code
                print(
                    f"\nStatus code: {response.status_code} \n")

                # Display the raw JSON data from the API
                print("The raw data from the API:")
                print(response.text)

                # Display the Python dictionary
                print("\nThe JSON data converted to a Python dictionary:")
                print(self._advice_data)

            # Display API data
            self.lbl_advice.config(
                text=self._advice_data.get("slip")["advice"])

        else:
            self.lbl_advice.config(
                text=f"API Unavailable: {response.status_code}")

# ------------------------------- CLOSE WINDOW ---------------------------#
    def close_window(self, *args):
        self.root.destroy()


advice_gui = AdviceGUI()
```

Example run:



Images

Labels and buttons (and other widgets) can display images instead of text.

To use an image requires a little set-up work. Create a **PhotoImage** object and give it a name. Here is an example:

```
cheetah_image = PhotoImage(file = 'cheetahs.gif')
```

Here are some examples of putting the image into widgets:

```
label = Label(image = cheetah_image)
button = Button(
    image = cheetah_image,
    command = cheetah_callback()
)
```

You can use the `configure` method to set or change an image:

```
label.configure(image = cheetah_image)
```

One unfortunate limitation of Tkinter is the only common image file type it can use is GIF. If you would like to use other types of files, a good solution is to use the Python Imaging Library.

Python Imaging Library

The Python Imaging Library (PIL) contains useful tools for working with images.

Install Pillow

Open a command prompt.

```
pip install Pillow
```

Using images other than GIFs with Tkinter: Tkinter can't use JPEGs and PNGs. But it can if we use it in conjunction with the PIL. Here is a simple example:

```
from tkinter import *
from PIL import Image, ImageTk
root = Tk()
cheetah_image = ImageTk.PhotoImage(Image.open("./assets/cheetah.jpg"))
button = Button(image = cheetah_image)
button.grid(row = 0, column=0)
mainloop()
```

The first line imports Tkinter. The next line imports a few things from the PIL. We load an image using a combination of two PIL functions. We can then use the image like normal in our widgets.

Assignment Submission

1. Attach the pseudocode or create a TODO.
2. Attach all tutorials and assignments.
3. Attach screenshots showing the successful operation of each tutorial program.
4. Submit in Blackboard.